

Load Interconnection Request System Impact Study Report

For Queue Project Q22-2206

Crystal River East 100 kV
March 15, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2206. The Interconnection Customer proposes a 79.4 MW large load (data center) facility interconnecting at the Crystal River East 100 kV substation in the Ohio area. The request was received on October 13, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 611 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$32,764,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
DYER to SILVER BOW 100kV L	IMPERIAL to EUFAULA 500kV	187.3	132	7.1
DE QUEEN to NORFOLK 100kV	EFAULA to IMPERIAL 500kV	156.2	198	7.1
SILVER BOW to JAMES RIV POWER STA 100kV line	EUFAULA to IMPERIAL 500kV L	150.6	112	11.4
DE QUEEN to HARTSVILLE 100kV	EUFAULA to IMPERIAL 500kV Line	134.7	255	7.1
GETTY to IX CENTER 100kV	EUFAULA SUBSTATION to IMPERIAL 500kV	133.6	260	5.1
CEDAR T to ABBE 100kV #1	ABBE to CEDAR PT 100kV Circuit 2	129.0	189	6.5
DYER to NORFOLK 100kV	TORO NEGRO to NORFOLK 100kV	119.2	126	6.6
POUST to FOX 100kV	EUFAULA SUBSTATION to IMPERIAL 500kV	117.9	241	8.5
COSUMNES to FOWLES 100kV Line	IX CENTER to GETTY 100kV L	117.8	180	6.7
ABBE to CEDAR POINT 100kV	ABBE to CEDAR PT 100kV	115.6	210	6.5
W UNION to WELLINGTON 1 100kV	EUFAULA to IMPERIAL 500kV L	113.1	251	8.0
NORFLK to TORO NEGRO 100kV Line	NORFOLK to DYER 100kV	111.5	123	5.9
CEDAR POINT to HARTSVILLE 100kV	HARTSVILLE to CEDAR PT 100kV Ckt 1	111.4	198	6.5
COSUMNES to GETTY 100kV line	GETTY SUBSTATION to IX CTR 100kV	110.4	221	6.7
WEST UNION to UNC 100kV line	HARTSVILLE to DE QUEEN 100kV	110.4	115	8.5
WELLINGTON 1 to FOWLES 100kV	IMPERIAL to EUFAULA 500kV	109.7	294	8.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
DYER to TORO NEGRO 100kV L	DYER to NORFOLK 100kV	109.7	96	5.9
JAMES RIV POWER STA to HUNTERS PT 100kV L	IMPERIAL to EUFAULA 500kV	108.3	127	11.4
CEDAR POINT to HARTSVILLE 100kV Ckt 1	HARTSVILLE to CEDAR PT 100kV	107.4	205	6.5
RIHARDSVILLE to LORING POWER PLANT 100kV	IMPERIAL to EUFAULA 500kV L	101.9	115	19.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DYER to SILVER BOW 100kV L	Rebuild line with higher-capacity conductor	\$2,129,000	100% (\$2,129,000)
DE QUEEN to NORFOLK 100kV	Rebuild line with higher-capacity conductor	\$2,942,000	100% (\$2,942,000)
SILVER BOW to JAMES RIV POWER STA 100kV line	Rebuild line with higher-capacity conductor	\$1,435,000	100% (\$1,435,000)
DE QUEEN to HARTSVILLE 100kV	Rebuild line with higher-capacity conductor	\$3,053,000	100% (\$3,053,000)
IX CTR to GETTY 100kV	Rebuild line with higher-capacity conductor	\$3,874,000	9% (\$340,000)
CEDAR T to ABBE 100kV #1	Rebuild line with higher-capacity conductor	\$2,971,000	100% (\$2,971,000)
DYER to NORFOLK 100kV	Rebuild line with higher-capacity conductor	\$1,704,000	100% (\$1,704,000)
POUST to FOX 100kV	Rebuild line with higher-capacity conductor	\$3,264,000	11% (\$349,000)
FOWLES to COSUMNES 100kV	Rebuild line with higher-capacity conductor	\$3,470,000	13% (\$436,000)
ABBE to CEAR PT 100kV Line 2	Rebuild line with higher-capacity conductor	\$1,252,000	100% (\$1,252,000)
WELLINGTON 1 to WEST UNION 100kV Line	Rebuild line with higher-capacity conductor	\$1,585,000	26% (\$416,000)
NORFOLK to TORO NEGRO 100kV	Rebuild line with higher-capacity conductor	\$2,895,000	100% (\$2,895,000)
CEDAR POINT to HARTSVILLE 100kV	Rebuild line with higher-capacity conductor	\$3,124,000	100% (\$3,124,000)
COSUMNES to GETTY 100kV line	Rebuild line with higher-capacity conductor	\$1,576,000	13% (\$198,000)
UNC to W UNION 100kV L	Rebuild line with higher-capacity conductor	\$5,643,000	100% (\$5,643,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WELLINGTON 1 to FOWLES 100kV	Reconductor line	\$587,000	26% (\$154,000)
TORONEGRO to DYER 100kV L	Reconductor line	\$698,000	100% (\$698,000)
JAMES RIVER POWER STATION to HUNTERS PT 100kV line	Reconductor line	\$1,329,000	100% (\$1,329,000)
CEDAR POINT to HARTSVILLE 100kV Ckt 1	Reconductor line	\$882,000	100% (\$882,000)
RICHARDSVILLE to LORING POWER PLANT 100kV Line	Reconductor line	\$1,381,000	59% (\$814,000)
	Total allocated to project		\$32,764,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.